# Introduction to Probability Theory

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February 21, 2024

# Outline


1. Probability Space
2. Axioms of Probability
3. Stochastic Processes
4. Brownian Motion and Martingales
5. Stochastic Calculus
6. References

# Probability theory is ubiquitous

We all hear the word probability (or chance, likelihood) in our day to day life for example

- There is a **high chance** that it will rain tomorrow.
- The party $X$ is **more likely** to form the government.
- A doctor assess the probability of successes of various alternative remedies.
- A bank wants to calculate the probability a loan will be repaid.
- A drug company calculates the probability of harmful side effects of drugs.

# Why Probability ?

The probable is what usually happens.

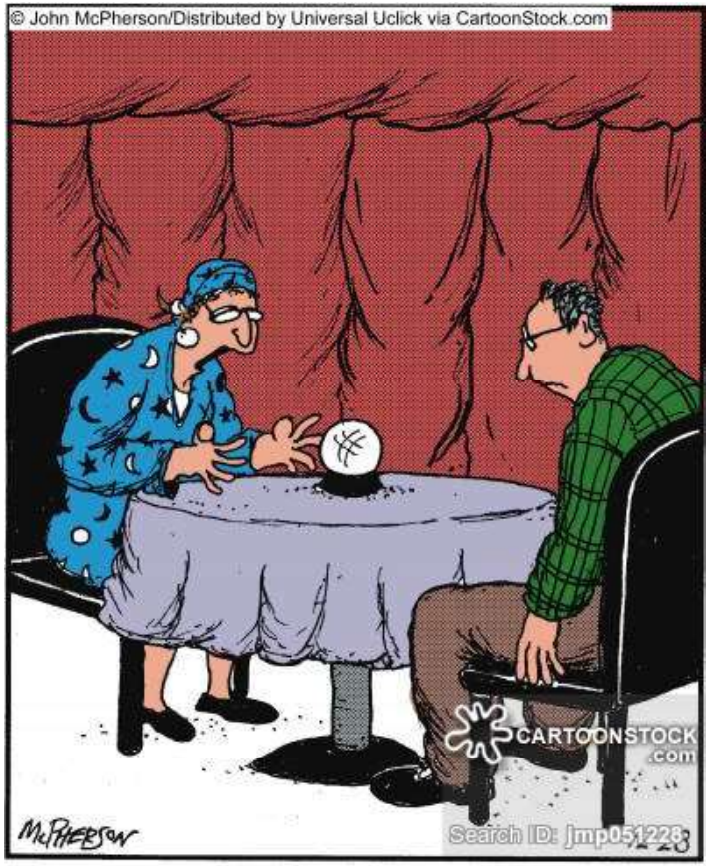
*Aristotle*

... it is a truth very certain that, when it is not in our power to determine the most true opinions we ought to follow the most probable.

*Descartes*

# Why Probability ?

From time immemorial human beings have been fascinated by the notion of looking into the future - probably because such a capacity would brings them certain advantages.



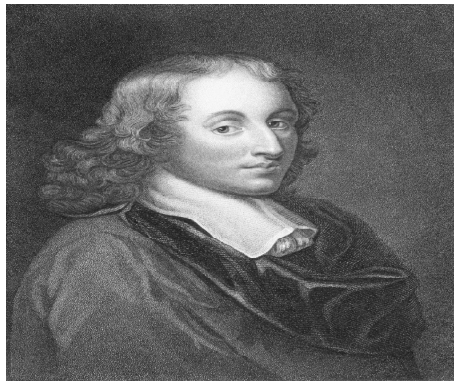

"I see a woman with a very large, round nose ... Oh, wait, that's my reflection. Let me try that again."

# Early work on probability

(a) Cardan (1501 - 1576)

(b) Kepler (1571 - 1630)

(a) Galileo (1564 - 1642) and Merton



(b) Pascal (1623 - 1662)

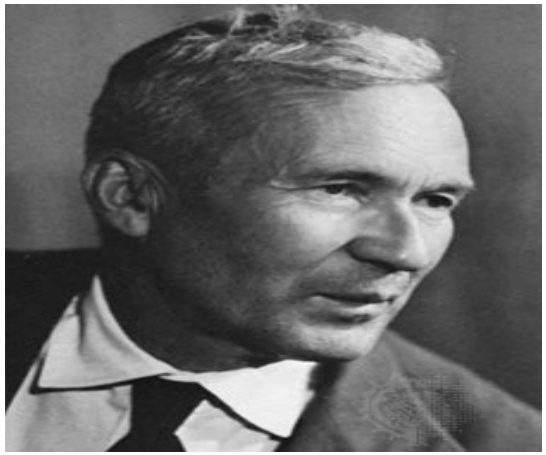
(c) Fermat (1601 - 1665)

# de Méré's Problem

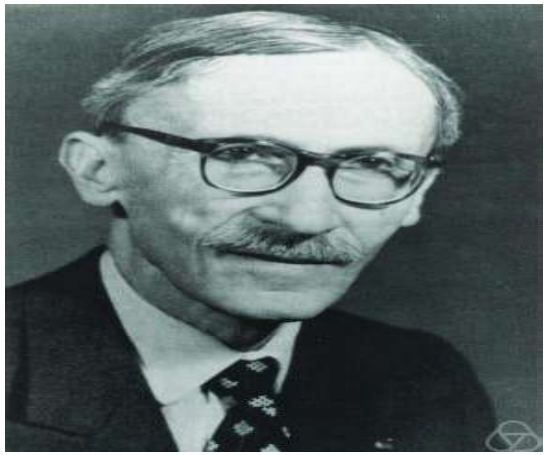
- The probability of getting at least one 'ace' in four roll of a single 6-sided die is higher than the probability of at least one 'double ace' in 24 throw of two dice.
- He posed this question to Pascal who took the help of Fermat to solve this.
- de Méreé came to the answer based on his practical experience and was not sure.

# Modern Probability

- In the twentieth century it was Kolmogorov (1903 - 1987) who saw the connection between the ideas of Borel and Lebesgue and probability theory.
- He gave the probability theory its rigorous measure theory basis.
- After the fundamental work of Kolmogorov, Paul Lévy (1886-1971) set the tone for modern probability with his seminal work on Stochastic Processes, Characteristic Function and Limit Theorems.



**(a)** Kolmogorov (1903 - 1987)



**(b)** Lévy (1886 - 1971)

### Sample Space

In a random experiment a sample space is a collection of all possible outcomes.

- The sample space is generally denoted by $\Omega$.
- The outcomes must be mutually exclusive.
- The outcomes must be collectively exhaustive.

### Example

- In a coin toss $\Omega = \{H, T\}$
- In a die roll $\Omega = \{1, 2, 3, 4, 5, 6\}$
- $\Omega = [0, 2\pi]$ if we are interested in angle of rotation in a wheel.

# The Events

- An **event** is a property which can be observed either to hold or not to hold after the experiment is done.
- An event is a subset of $\Omega$.
- For two events $A$ and $B$, the event "*A or B*" is interpreted as the union $A \cup B$.
- The contrary event is the complement set $A^c$.
- The event "*A and B*" is interpreted as the set $A \cap B$.
- The impossible event is the empty set $\phi$
- An **elementary event** is a "singleton" i.e. $\{\omega\}$.

# Axioms of Probability

**Definition (Algebra)**

Let $\Omega$ be a non-empty set, and let $\mathcal{F}$ be a collection of subsets of $\Omega$. We say that $\mathcal{F}$ is a an algebra provided that:

i. $\Omega \in \mathcal{F}$.

ii. whenever a set $A \in \mathcal{F}$, its complement $A^c$ also belongs to $\mathcal{F}$.

iii. if $A_1, A_2, \ldots A_n \in \mathcal{F}$ then $\cup_{i=1}^{n} A_i \in \mathcal{F}$.

**Example**

- $\mathcal{F} = \{\phi, \Omega\}$
- $\mathcal{F} = \{\phi, \Omega, A, A^c\}$.

**Definition ($\sigma$-algebra)**

Let $\Omega$ be a non-empty set, and let $\mathcal{F}$ be a collection of subsets of $\Omega$. We say that $\mathcal{F}$ is a $\sigma$-algebra provided that:

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ii. whenever a set $A \in \mathcal{F}$, its complement $A^c$ also belongs to $\mathcal{F}$.

iii. if $A_1, A_2, \ldots A_n, \ldots \in \mathcal{F}$ then $\cup_{n=1}^{\infty} A_n \in \mathcal{F}$.

**Example**

- $\mathcal{F} = \{\phi, \Omega\}$
- $\mathcal{F} = \{\phi, \Omega, A, A^c\}$.

**Example**

Let $\mathcal{F}$ be a $\sigma$-algebra on $\Omega = [0,1]$ such that $[\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$ for $n = 1, 2, \ldots$. Show that

a. $\{0\} \in \mathcal{F}$
b. $\{\frac{1}{n} : n = 2, 3, \ldots\} \in \mathcal{F}$.
c. $(\frac{1}{n}, 1] \in \mathcal{F}$ for all $n$
d. $(0, \frac{1}{n}] \in \mathcal{F}$ for all $n$.

We have

a. $\{0\} = [0,1] \setminus \cup_{n=1}^{\infty} [\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$.
b. $\{\frac{1}{n} : n = 2, 3, \ldots\} = \cup_{n=2}^{\infty} \left( [\frac{1}{n+1}, \frac{1}{n}] \cap [\frac{1}{n}, \frac{1}{n-1}] \right) \in \mathcal{F}$.
c. $(\frac{1}{n}, 1] = \cup_{i=1}^{n-1} [\frac{1}{i+1}, \frac{1}{i}] \setminus [\frac{1}{n+1}, \frac{1}{n}] \in \mathcal{F}$.
d. $(0, \frac{1}{n}] = \cup_{i=n}^{\infty} [\frac{1}{i+1}, \frac{1}{i}] \in \mathcal{F}$.